The Smell of Knowledge: Attenuating Variety in the Age of AI

A Viable Fork

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Abstract

AI generation is outpacing mathematics' verification infrastructure: the bottleneck is the human "smell" Klowden and Tao (2026) identify. This paper diagnoses the bottleneck as a requisite-variety failure (Ashby, 1956): generation variety has overtaken regulator variety, and the institutional response must attenuate one and amplify the other. The proposed architecture relocates the bottleneck across existing institutions: proposers formalize and submit under honestly declared axioms, repositories organize and map the verified record, and journals curate for their readership. The architecture yields a topological object over verified mathematics, available for meta-mathematical analysis, and positions mathematics to serve as shared structural infrastructure across the empirical sciences at a moment when every discipline's verification arrangements are under requisite-variety pressure.

1 The Sommelier and the Two Crises

1.1 The Nose

A sommelier opens a bottle of wine to pour a guest a glass. She lifts the glass to her nose, does not taste, does not consult the technical sheet, does not question the wisdom of ordering this glass. She inhales, and in that fraction of a second, a trained perceptual system cross-references the wine's volatile profile against thousands of prior encounters. Tonight, the signal is faint but unmistakable: 2,4,6-trichloroanisole. Cork taint. The wine is corked. She pulls the bottle before it reaches the bar. A guest with an untrained nose would have tasted it, found something vaguely off, perhaps blamed the food pairing or their palate. A chemical assay would have confirmed the TCA contamination, but only after the bottle had been poured. The sommelier's nose operates in the gap between sensation and analysis: it catches what formal testing would eventually confirm, but catches it *before the wine is served*.

This is one of three things she does. Custodial fault detection on bottles where no guest holds proofing authority is the most institutionally visible function, the part the establishment depends on her for. Upstream, off the floor, she curates the list: selecting wines that cohere with the establishment's identity, balancing producers and vintages and price points against the kitchen's program. And in dialogue with the guest, she recommends and educates: she listens to a stated preference, often one the guest cannot

fully articulate, translates it into the vocabulary of the list, and guides the guest toward a bottle that will satisfy or expand their palate. Three functions, performed by one trained perceptual system, organized through the institutional structure of the restaurant. This paper argues that mathematical practice currently confuses these three functions in a way the AI moment makes untenable, and that the architecture proposed here separates them institutionally while letting mathematics, under the architecture, perform all three at the level of disciplinary infrastructure.

Klowden and Tao (2026), in their examination of AI's impact on mathematical practice, identify an analogous capacity in experienced mathematicians. They call it the "smell test": a pre-theoretical assessment of whether a proposed proof carries the markers of genuine mathematical understanding, executed before line-by-line verification begins. They ground this capacity in Thurston's (2006) account of what a proof contributes beyond formal correctness: the narrative structure that shows why the argument works, which steps do the heavy lifting, how the result connects to the broader mathematical landscape. The smell test is the trained perception of this Thurstonian content, or of its absence. The experienced mathematician detects when an argument lacks causal narrative, when conclusions arrive without the structural work that should support them, when the proof is, in a word borrowed from Klowden and Tao's own framing, *odorless*.

1.2 The Two Crises

Klowden and Tao's "odorless proof" bundles two distinct failure modes under a single label. The first is a *verification crisis*: proofs that contain hidden structural errors camouflaged by surface aesthetics. These are corked bottles. The wine is flawed; the flaw is concealed by an attractive label and confident service. The sommelier's job is to catch them.

The second is an *interest crisis*: proofs that are genuinely correct, pass every formal test, and contribute nothing to the field's self-understanding. These are sound but vacant wines. No TCA, no volatile acidity, no brettanomyces, correct pH, correct residual sugar, and yet: no complexity, no development, no finish. The wine exists on the spec sheet. It is absent in the glass.

The distinction matters because the two crises require fundamentally different infrastructure. The verification crisis is an engineering problem: it asks for a mechanical system that catches structural faults before flawed proofs enter the institutional record. The interest crisis is a curatorial and pedagogical problem: it asks for human judgment operating through citation, peer review, pedagogy, and paradigm formation, the slow social processes by which the mathematical community decides which correct results are worth remembering. Asking a mechanical verification system to evaluate whether a correct proof is *interesting* is like asking the sommelier at the bar to critique the guest's choice of bottle. The jurisdiction is wrong.

1.3 Thesis and Plan

This paper diagnoses the bottleneck Klowden and Tao identify as a requisite-variety failure and specifies the institutional architecture that resolves it. Ashby's (1956) law of requisite variety states that a regulator can govern a variable only when the regulator's variety meets or exceeds the variety of the variable being governed. Mathematical practice has historically maintained a regulator (the trained smell of experienced practitioners, the journal review process, the citation network) whose variety was sufficient for the historical generation rate of mathematical content. AI generation overtakes this regulator, and the institutional response must close the variety gap. The architecture closes the gap through both directions of Ashby's principle: it attenuates the variety the regulator must absorb (mechanizing structural-fault detection, clustering structurally identical proofs, surfacing axiom-dependency structure) and it amplifies the regulator's own variety (institutionalizing tacit field-knowledge in the queryable atlas, scoping foundational revisions through the dependency chart, supplying coordination signals to the volunteer formalization effort). Which variety to attenuate and which to amplify is a substantive choice rather than a cybernetic deduction; the architecture's choices reflect a particular reading of where mathematical practice currently fails and where it could grow. The architecture's three layers map onto the sommelier's three functions: the verification layer mechanizes custodial fault detection through dependent type theory; the institutional layer reorganizes repositories around the mechanized verification function and frees journals to specialize in curation; the pedagogical layer maintains the human capacity that lets the atlas serve as recommendation infrastructure for users approaching mathematics from outside their own expertise. The three layers rest on one architectural principle: the onus of formalization sits with the generator, discharged modularly against honestly declared axioms.

One definition carries the architecture and is stated here because the onus principle of Section 3 follows from it rather than from convenience. A claim counts as knowledge to the degree it survives attenuation of its producer: turn down reliance on who is speaking and how well they present, and knowledge is what holds still. A proof runs this to completion, since checking the steps removes the author entirely. A measured constant runs it as far as independent replication reaches, since what survives is that separate measurements agree. Both are the same operation performed to different depths. The definition is an invariance condition of the kind Klein's Erlangen program gave geometry, with the producer as the parameter varied and standing as the invariant, and it sorts the two crises above along the line already drawn: a vacant proof survives producer-attenuation completely and remains uninteresting, so the definition speaks to the verification jurisdiction and stays silent on curation. A definition with opinions about interest would commit the jurisdiction error of §1.2. This one declines to.

The paper proceeds in three parts. Section 2 introduces the constructs required for the verification layer: a revisable constraint hierarchy, a record of what has been structurally eliminated, and a recursive protocol for revising the hierarchy itself. Section 3 states the onus principle, introduces the modular discipline that makes it deployable, and describes the theory atlas that emerges as the positive record of verified knowledge. Section 4 specifies three commitments that constitute the verification layer and addresses the objection that voluntary adoption of the onus is unstable. Section 5 works out the

institutional consequences for mathematics: repositories become verifiers, cartographers, and coordinators; journals specialize in curation; pedagogy preserves the human capacity the architecture presupposes. Section 6 extends the consequences beyond mathematics itself: the architecture's output is an institutional resource for the empirical sciences more broadly, and mathematics, under the architecture, is positioned to take the institutional lead in organizing the shared structural ground across disciplines that will face their own requisite-variety crises as AI reaches their verification infrastructures.

One distinction must be drawn early. The architecture equilibriates what we will call Game 1: the distributed search for mathematical truth, whose players are generators producing candidate proofs, verifiers (proof-assistant kernels) checking them, and the research community maintaining the institutional infrastructure. Mathematical practice also involves Game 2: the allocation of credit, priority, publication, archival authority, and institutional reward. Game 2 has a different payoff structure and different equilibria.

The architecture partitions Game 2 rather than declining it, and the partition follows from the definition above. Certify what recomputes from public structure, and leave free what does not. The discriminator is recomputation because a mechanism that recomputes removes a target, since anyone can rerun it and no position on it is worth capturing, while a mechanism that is fixed without recomputing installs one. Both halves of the argument the paper runs against trust-in-producer apply here in the same form. A fixed measure of what mathematical work is worth is a fixed signal that recursive modelers aim at, so it is manufactured more cheaply than the substance it certified, which is the documented career of the impact factor and the h-index. And a single standard of what matters is a low-variety regulator set against a plural field, which is the jurisdiction error of §1.2 committed at the level of reward rather than of verification.

Stated as a budget, which is the form that makes the partition a design principle rather than a boundary: certification is the cheap resource and judgment is the scarce one, so the architecture certifies everything certifiable in order that judgment is spent only where nothing else can be spent. The field's judgment is the common-pool resource this architecture manages. It is subtractable, since reviewer attention consumed on a structural check is unavailable for a curatorial one, and its over-appropriation is the crisis §1.2 opens with. Ostrom's (1990) design principles for long-enduring institutions governing such resources supply the empirical record the partition draws on, and Section 5 works out where each principle lands.

The partition assigns unevenly, which is what makes it a claim. Priority recomputes: admission is dated, proof terms are content-addressed, and the reference atlas of §5.3 computes the anticipation neighborhood from content rather than from what an author declares. Priority therefore sits in the certified tier and the architecture specifies it. Attention, prestige, and institutional reward recompute from nothing, and they are the site of the plural judgment the field exists to exercise, so they sit in the free tier and the architecture specifies no equilibrium for them. That silence is the position rather than the omission. The claim is falsifiable in the form any thesis of this kind should be: a fixed mechanism for

allocating attention that survives a generation of practitioners modeling it would refute the partition, and the metric record is the evidence the paper offers that none has.

2 Institutional Memory and the Compost Pile

2.1 Science as Distributed Contraction

A hypothesis is a structured conjecture expressed with enough precision to admit severe testing. Verification returns evidence that updates the collective posterior; falsification returns a permanent structural datum about what does not work. Pure mathematics performs this at maximum severity: a proof closes an entire class of alternatives, and the closure is irreversible within the axiomatic framework that produced it. Across instances, the discipline's work is the distributed contraction of the space of viable hypotheses.

The engine's power derives from continuity across time. A single researcher operates within a career spanning decades. The institution extends that horizon indefinitely. Institutions carry the accumulated negative knowledge, the structured memory of what has been tried and killed, that prevents the engine from resetting to its uninformed prior with each new generation. Without this memory, the engine suffers from *zombie iteration*: the indefinite recurrence of previously eliminated approaches. The pharmaceutical industry's file drawer problem (Rosenthal, 1979) illustrates the pathology. NASA's Lessons Learned Information System illustrates the remedy.

Popper (1972) argues that objective knowledge accumulates as World 3: the growing body of problems, theories, and critical arguments whose existence is independent of any individual knower. The accumulation proceeds through elimination. Theories are proposed, tested, falsified, and what survives constitutes the current best-accumulated conjectural content. The architecture specifies a specific institutional form for World 3 in mathematics: a structured, queryable record of demonstrated structural impossibilities (the Compost Pile below), paired with a verified library whose structural topology is cartographically legible (the theory atlas, Section 3). Both sides of Popperian accumulation, the negative record of eliminations and the positive record of what has been established, require operational form if they are to survive the acceleration AI generation introduces.

2.2 The Trellis: A Revisable Constraint Hierarchy

The *trellis* is a tiered hierarchy of constraints $\Sigma = \{C_0, C_1, \ldots, C_n\}$ equipped with a partial order representing logical strength and defensibility. Constraints at higher tiers are more defensible and harder to revise; constraints at lower tiers are more specific and more readily revisable. A candidate hypothesis is *viable relative to* Σ if it satisfies every constraint in Σ. In its strongest instantiation, each constraint corresponds to a typeclass in a dependent type theory (de Moura and Ullrich, 2021), and a hypothesis satisfies a constraint when it provides a proof term inhabiting the required type.

The trellis extends Lakatos's (1970) distinction between a hard core of foundational commitments and a protective belt of mutable auxiliary hypotheses, with two modifications. First, the hard core's protection is conjectural rather than conventional: every constraint is defended on evidentiary grounds and revisable when the evidence changes (against Lakatos, who treats the core as protected by methodological decision). Second, the hierarchy admits formalization: in a proof assistant, trellis constraints become typeclasses, and hypothesis testing becomes compilation.

The trellis is self-adopted by a research group, owned by the practitioners who maintain it. This is a structural feature: a group that games its own best practices degrades only its own work, whereas a group gaming an externally imposed standard extracts a private benefit at public cost. Self-adoption resolves the motivated-revision problem by making the problem self-defeating. The architecture works because practitioners with the judgment to maintain it hold the authority to revise it. Under the requisite-variety framing, the trellis is the institutional apparatus that supplies the regulator with explicit access to the variety of foundational commitments operative in the field; tacit foundational commitments carry variety the institutional regulator cannot govern, and the trellis closes that gap by making the commitments declarable.

2.3 The Compost Pile

The *Compost Pile* is an append-only database that records every demonstrated structural impossibility. Each entry records the blocked hypothesis, the blocking constraint, the mechanism of failure, the block scope (point or class), the structural residue that survived, the reactivation conditions (a predicate on the trellis), and the trellis version that produced the block. The full schema:

Field	Content	Example
Entry ID	Unique identifier	BLOCK-2026-003
Hypothesis	The mechanism tested	"Scalar gradient sources axial torsion"
Blocking Constraint	Which trellis tier blocked it	Tier 2: parity invariance
Mechanism	How it failed	Parity mismatch: true vector $\neq$ pseudo-vector
Scope	Point (specific) or Class (family)	Class: all true-scalar-gradient-to-pseudo-vector couplings
Residue	What survived	Trace channel (scalar QFI density, no gradient coupling)
Reactivation	When the block dissolves	Demotion of Tier 2 parity invariance
Trellis Version	Which Σ_k produced the block	Σ_3

A *query* against the Compost Pile retrieves all entries whose hypothesis is structurally similar to a candidate, or whose block mechanism identifies the pattern the candidate instantiates. The query returns

either an empty set (no known anti-pattern) or the specific structural reasons why similar approaches have previously failed, together with the conditions under which the block would dissolve.

The Compost Pile instantiates Popper's (1972) World 3 in a specific operational form: a structured, queryable repository of demonstrated impossibilities. The information-theoretic value of a blocking result follows directly from treating the viable set as a uniform prior over the $|T|$ remaining candidates: eliminating one reduces the Shannon entropy by $K_{\text{null}} = \log_2 |T| - \log_2 (|T| - 1) = \log_2 (|T|/(|T| - 1))$, which grows as $|T|$ shrinks, so each block is worth more the closer the search is to resolution. The Compost Pile amplifies the institutional regulator's variety by making accumulated negative knowledge externally accessible, where it had previously lived in the tacit memory of senior practitioners and the scattered record of failed projects.

2.4 The Shore: Recursive Revision

The *Shore* is a periodic reassessment protocol that examines the trellis's own constraints and asks whether they accurately represent the structural knowledge they encode. When a constraint is demoted, the trellis is modified and the Compost Pile is queried: entries whose sole blocking constraint has been weakened re-enter the viable set. When a constraint is promoted, previously viable hypotheses may be blocked. The Shore operates at the meta-level (updating the trellis) while the adversarial audit operates at the object level (testing hypotheses against a fixed trellis).

The name carries an image. A research program beaches its vessel periodically to inspect the hull, scrape accumulated barnacles, and ask whether the vessel remains seaworthy. In viticulture, the analog is the dormant-season assessment: the grower walks the vineyard after harvest, evaluates the trellis system, prunes dead wood, and decides whether the training architecture still serves the vineyard's evolving conditions.

The Shore gives the Kuhnian (1962) crisis an institutional form. Where Kuhn describes paradigm shifts as episodic reorganizations following the accumulation of anomalies too severe for a paradigm to accommodate, the Shore schedules the reassessment at the meta-level of the research program and makes trellis revision a continuous evidence-driven function. The Compost Pile's records survive the Shore with full metadata, so revision operates on an explicit rather than tacit record of what the predecessor trellis blocked. The Shore instantiates System 5 of Beer's (1972) Viable System Model: the recursive subsystem that reassesses foundational commitments when environmental signals indicate they have become maladaptive. The Shore's signals are quantitative, drawn from the Compost Pile's metadata: which constraints block the most hypotheses, which blocks are class-level, which reactivation conditions recur. Under the requisite-variety framing, the Shore is the temporal calibration mechanism: as the variety of substantive content shifts over time, the regulator's structure must shift to maintain match, and the Shore performs that recalibration on evidence-driven schedule.

2.5 Monotonicity and Revision

A potential tension arises between two claims. First: within a fixed trellis, the viable set contracts monotonically under sustained adversarial audit. Second: the Shore can revise the trellis itself, so previously blocked hypotheses may re-enter the viable set after a Shore review.

The resolution distinguishes object-level and meta-level operations. Within any fixed trellis Σ_k, contraction is monotonic: $V_{t+1}(\Sigma_k) \subseteq V_t(\Sigma_k)$ for all t. Shore reviews operate at the meta-level, updating the collective posterior over competing trellis hypotheses. When evidence shifts the posterior toward a trellis with weakened constraints, previously blocked hypotheses may re-enter: $V_0(\Sigma_{k+1}) \not\subseteq V_t(\Sigma_k)$. This is model selection over trellis hypotheses. Within each version, contraction is strict. Across versions, the Compost Pile preserves every block with full metadata, enabling reconstruction of the contraction history under any past trellis geometry.

The Shore's scope scales with the tier of the revised constraint. A revision at a low tier, for instance tightening a regularity assumption that only a narrow portion of the library depends on, triggers minor reformalization. A revision at a higher tier, for instance a Tier 2 promotion that changes what proofs must satisfy to remain admitted, triggers proportionally broader consequences. A revision at the highest tiers, for instance demoting the law of excluded middle in a move to constructive foundations, is what Kuhn would call revolutionary change: large portions of the library require reformalization against the new foundation, and the axiom-dependency chart specifies exactly which portions. The Shore is one mechanism, and the magnitude of its intervention scales with the tier it touches. The distinction between normal and revolutionary change, under the architecture, is quantitative rather than categorical. The philosophical implications of this collapse, particularly its relationship to Kuhnian incommensurability and to Ashby's law of requisite variety, are taken up in §6.5.

3 The Onus of Formalization

3.1 The Principle

The architectural principle underlying the verification layer: whoever wants a claim admitted to the verified record must produce a type-checking proof term. The cost of formalization sits with the generator, not with the verifier.

The principle follows from the definition in Section 1. If standing is what survives attenuation of the producer, the record's admission criterion is whatever performs that attenuation, and a type-checking proof term performs it completely: the kernel's verdict is a function of the term and the ambient theory, and it holds fixed across every variation in who submitted it. The onus is the definition made operational at the point of admission, which is why it is available as a requirement rather than as a request. A generator asking for admission is asking the record to carry a claim whose standing anyone can recompute, and the proof term is what makes the recomputation possible.

This inverts the current distribution of verification labor in mathematical publishing. Authors currently submit informal proofs in natural language; reviewers bear the cost of checking informal argumentation; structural flaws can persist when reviewer effort is bounded. Under the onus principle, a generator's claim does not enter the record until it compiles. The verifier's cost is mechanical and tractable. The generator's cost is non-trivial and borne before submission.

Verification's cost migrates to the side of the transaction that wants the admission. Once the verifier's cost is cheap, the institutions performing verification can absorb the function at scale, and the institutions that previously verified (journals) are freed to specialize in the work they do better than any mechanical system. Under the requisite-variety framing, the onus is variety migration: the proposer was always the source of the variety the regulator was attempting to match, and the onus aligns the institutional structure with that fact. The proposer carries the variety to their submission as a compiled proof term; the regulator's variety requirement collapses to mechanical type-checking, which scales with computational capacity rather than with bounded human attention.

The onus extends to documentation. A type-checking proof term is a necessary condition for admission but not sufficient. The formalization community's existing practice, visible in Mathlib's style conventions (The mathlib Community, 2020), requires that proofs be accompanied by descriptive lemma names, docstrings stating what each result says and sketching the approach, comments at strategic points explaining why particular tactics are applied, and references to adjacent work in the library. These artifacts carry what Thurston (2006) calls the narrative structure of a proof: which steps do the heavy lifting, which are routine, how the argument fits into the broader literature. A proof whose internal structure is opaque (Klowden and Tao's concern, instantiated in AI-generated outputs that pass type-checking while containing redundant or inexplicable steps) fails admission under this extended onus for the same reason it would fail an experienced human referee: the content required for downstream use is missing. The repository's admission is a gate against both verification failure and documentation failure.

3.2 Modular Axioms and Conditional Formalization

The onus principle, naively interpreted, is undeployable. Requiring every admitted proof to compile against a fully formalized library means waiting decades for Mathlib or an analog to be complete. The *modular axiom* discipline solves this. The generator's formalization obligation is to produce a type-checking proof conditional on honestly declared axioms, where the axioms stand in for accepted-but-unformalized prior results.

Each declared axiom carries provenance: the informal theorem it represents, the textual source of that theorem, the scope and signature of the claim the axiom encodes, and a status flag (undischarged, discharged, retracted). The axiom is a formal object visible in the repository's records. When the referenced theorem is formalized by the broader community's formalization effort (Lean's Mathlib, analogous efforts for other proof assistants), the axiom is discharged: the paper's proof now compiles

against the formalized theorem rather than the placeholder. If a discharge reveals the axiom was misstated, the paper is flagged for revision.

The discipline has three consequences. First, the architecture is deployable from day one. A paper on a specialized topic can enter the verified record now, conditional on axioms that Mathlib will formalize over decades. Second, the collective set of declared axioms constitutes a demand signal for the formalization community, surfacing which unformalized results are actually in use and prioritizing the effort accordingly. Third, the axiom-declaration discipline is itself a form of epistemic hygiene: the generator is obligated to state honestly what they assume, and misstatements become visible when discharge is attempted. Modular axioms are the variety-attenuation move that makes the onus deployable: the proposer carries only the variety their proof actually depends on, declared explicitly, with the remaining variety standing in as honestly-flagged placeholders.

The discipline forces a distinction the paper's vocabulary has so far left implicit, and naming it converts what follows from a concession into a position. A conditionally formalized result carries two gradings on incomparable axes. That the theorem follows from the declared axioms is certified: the kernel checks it, and the certification is constitutive, holding by the structure of the derivation. That each axiom statement is faithful to the informal theorem it stands for is empirical: it is checked when a formalization attempt discharges it, and independently rechecked when a second attempt from a different context discharges it again. The two axes admit no comparison. A result at the constitutive position on the first says nothing about its position on the second, which is why a fully compiled paper resting on a misstated axiom is a coherent object rather than a contradiction, and why the record displays both positions rather than one summary grade. Mathematics is distinctive in how far the first axis reaches and it shares the second with every empirical discipline, because an axiom declaration is a claim about the literature and the literature is part of the world.

The discipline has an acknowledged failure mode. A declared axiom can be misstated. The system surfaces the misstatement on discharge rather than burying it, which is an improvement over the current practice of informal citation, but the paper does not claim axiom statements are automatically correct. Where multiple formalization attempts succeed against the same declared axiom, the repeated success provides accumulating evidence of axiom soundness to the degree the attempts are independent, and independence is a claim the record carries rather than a property the infrastructure infers. Each discharge attempt declares its footprint: the proof assistant, the ambient library version, the tactic set, the generator that produced the term, the author who wrote the statement. Two attempts sharing the whole footprint are one attempt performed twice, and the record admits them at that label rather than promoting them to corroboration. The closure over declared footprints is what separates genuinely independent discharges from shared ones, it is computable from the declarations alone, and it is what the second axis of the previous paragraph is graded against. Systematic misstatements of the kind Tanswell (2015) analyzes stay invisible until an attempt whose footprint overlaps little with the prior ones discharges the axiom again. Where researchers disagree about whether a declared axiom is stated faithfully or scoped correctly, the dispute resolves through the same Shore mechanism that handles other trellis revisions:

evidence accumulates through discharge attempts, the research community evaluates the evidence, and the axiom's scope or statement is revised if the evidence warrants.

3.3 Crossing Between Foundations

Mathematics has plural foundations, and an architecture that admits results into a record must say what happens when a result crosses between them. A theorem proved with excluded middle and the axiom of choice is a theorem of classical mathematics. Imported into a constructive repository, it is a well-formed object carrying its provenance and holding no local standing, because a repository's grades are computed against its own constraints and the classical proof satisfies constraints the constructive repository declined to adopt.

The rule the architecture adopts: an imported result grounds nothing until a local author types it under the receiving foundation and signs the typing as a claim of their own. The import is retained with provenance intact, so the crossing is auditable and the original is citable as exactly what it is. What the import lacks is the capacity to serve as support for anything downstream, which is the property that would otherwise let standing earned under one foundation propagate silently into a repository that never granted it. The cost of the crossing sits with whoever wants the crossing, which is the onus principle applied at a boundary rather than at an admission.

The paper already runs this move at a smaller grain. An undischarged axiom grounds the implication resting on it and grounds nothing about the referent until discharge. The crossing rule is the same operation performed at the boundary between repositories rather than at the boundary between the formalized and the unformalized, and the reason is identical in both cases: standing is computed locally from what local constraints certify, so anything arriving from outside enters as a declaration awaiting local work.

This bounds a failure mode Section 5 otherwise leaves open. A repository whose admission standards have decayed, whose atlas has been captured, or whose declared axioms have drifted from their referents exports results that confer standing nowhere else until some local author stakes their own name on retyping them. Capture stays contained to the repository that suffers it, and the containment holds from the first repository onward, without requiring the plurality of rivals that the competitive argument of §5.2 presupposes. This is a partial answer to the bootstrapping gate rather than a discharge of it. The gate concerns who builds the first atlas and what prevents its capture during formation; the crossing rule addresses only what a captured atlas can spread to, which leaves the formation question open and shrinks the blast radius of the answer being wrong.

3.4 The Theory Atlas: What Mathematics Can Do

The verified library and its axiom declarations carry structural information beyond the individual results they record. Dependency relations show which theorems use which. Typeclass and structure hierarchies

show which objects instantiate which abstractions. Technique clusters show which proof strategies recur across domains. Bridging patterns show where techniques from one area formally apply to another. Density patterns show where the library is thick and where it is thin. The axiom-dependency chart shows which results depend on which axioms, which axioms remain undischarged, which have been discharged by library formalization, which have been shown inconsistent and retracted.

The *theory atlas* is the structural-organizational object that aggregates these relations into a queryable topology over the library. Each relation is a chart; the atlas is the collection, compatible on overlaps. When a theorem appears in both the dependency chart and the technique chart, it is the same theorem, and a modification to its formal statement propagates to both. When an axiom is discharged by library formalization, every chart that referenced the axiom updates consistently. The atlas is a single object viewed from multiple angles, not several objects with nominal cross-references.

The atlas vocabulary extends existing practice in the formalization community. The term *library* is standard across Lean's Mathlib (The mathlib Community, 2020), Isabelle's Archive of Formal Proofs (Paulson, Nipkow, and Wenzel, 2019), Coq's standard library, and Mizar's Mathematical Library (Bancerek et al., 2018). The structural topology has been described under several partial terms: *typeclass hierarchy* in Mathlib, *theory graph* in Isabelle, *dependency graph* generically. Each picks out one chart. The atlas is the collection: it holds these partial views together as consistent charts of a single library, extended to include the axiom-dependency chart the modular discipline requires. The atlas metaphor follows the classical mathematical use of *atlas* as a collection of coordinate charts covering a manifold; the cartographic sense applies, and the technical apparatus of point-set topology sits elsewhere.

The atlas is constructed, not read off. Different repositories face a design choice about which charts to build and how to organize them, and the choice shapes which queries the atlas answers well. One possible organization is by mathematical subdomain, which mirrors how mathematics is traditionally taught and published. Another is by functional specification: what a transformation takes as input, what it produces, what it preserves, what it breaks. A third presents the library's contents under whatever organizational axis a user's query selects, with other axes available as metadata. Different repositories will make different choices, and the field's experience with the resulting atlases will determine which choices prove robust. Under the requisite-variety framing, the atlas is the variety amplification mechanism for the institutional regulator: the field's structural self-knowledge, formerly carried as the tacit variety of senior practitioners, becomes externally accessible variety that any agent (human or AI) can query.

Atlas construction carries a cost the verification layer does not, and the paper's claim that verification is mechanical and tractable does not extend to it without argument. Type-checking a submitted proof term is local: the kernel reads the term and the ambient theory and returns a verdict, and the cost scales with the submission rather than with the library. The atlas is global by construction. Dependency closure, technique clustering, bridging detection, and the computed anticipation neighborhoods of §5.3 are relations over the whole holding, and recomputing them across a growing library on every admission is

the arrangement that fails first as generation accelerates. The resolution is structural and follows the atlas metaphor rather than straining it. Charts are local by definition, and the atlas is the collection compatible on overlaps. Verification is therefore scoped to a region: an admission recomputes its own region's charts and the overlaps that region participates in, and the global atlas is the composition of verified regions rather than an object reverified as a whole. Regions whose overlaps are unaffected by an admission do not recompute. This bounds construction cost by the region rather than by the library, and it is what makes the cartography layer deployable at the rate admissions arrive rather than at the rate a full recomputation allows.

Regional scoping raises the question of who authors a region, and the answer is not a matter of operator convenience. The atlas is the field's self-knowledge about its own productive capacity, and a region authored centrally substitutes one group's model of a subfield for the subfield's own, which reinstates at the level of cartography exactly the tacit-variety problem the trellis was introduced to close. A region should be authored by the practitioners working inside it, whose sense of which techniques transport and which boundaries are real is the variety the region is meant to make explicit. Ostrom's Design Principle 8, nested enterprises for coordination problems too large for a single unit, is the structural form: regions are authored locally and compose at their overlaps, and the shared apparatus carries what composition across overlaps requires and no more.

3.5 Atlas as the Field's Self-Knowledge

Mathematicians have always had implicit knowledge of the discipline's productive capacity: which techniques transport, which regions connect, which problems share structure, where the dense corridors run and where the sparse ones do. This knowledge is Thurston's (2006) subject when he argues that mathematical progress depends on advances in understanding as much as in results. But Thurstonian understanding at the discipline-level has historically lived in the tacit memory of senior practitioners, distributed unevenly across communities, lost when practitioners retire or change fields, reinvented within subdisciplines that do not share vocabulary. The atlas makes this self-knowledge cumulative, queryable, and transmissible. Dependency structure records what has been built from what. Technique clustering records which methods have proven portable. Bridging structure records where the field's productive capacity extends across nominal boundaries. The atlas is a structural account of what mathematics does, derived from the verified library as its natural cartographic output. Questions of mathematical ontology, whether structures are discovered or constructed and what their existence consists in, sit elsewhere; the atlas records the productive topology. Section 6 offers a characterization of mathematics that turns on answerability rather than existence, and it is scoped there as a positioning claim the architecture holds independently of.

Philosophy of mathematics has long had to distinguish what an individual mathematician knows from what the discipline knows as a collective subject. The standard answer to the collective question points to the literature plus the tacit training that practitioners undergo, and treats the combination as the field's knowledge. The atlas offers a more specific answer: the discipline's self-knowledge about its own

productive capacity has a specifiable institutional form, derivable from the verified library by rules, maintainable by repositories, and queryable by any agent with access to the repository's holdings. The collective subject has an institutional object it can consult.

AI generation produces proposals: candidate proofs, candidate axiom declarations, candidate technique applications. The Linter is one component of the acceptance criterion, evaluating structural soundness. The smell is the other component, evaluating direction: which proposals are worth pursuing further, which bridges are likely to be productive, which regions of the landscape the field's accumulated judgment identifies as worth exploring. Without the smell, the chain explores uniformly within the space of correct proofs, which is enormous and uninteresting. With the smell, the chain is biased toward regions the field's self-knowledge marks as dense, productive, or unexplored. The relationship between mechanical verification and human judgment has the structure of a Markov Chain Monte Carlo sampler, with the Linter and the smell as the two acceptance components and the atlas as the externalized representation of the smell biasing the chain across submissions. Both components are required. The Linter alone produces a flood of correct content without direction; the smell alone has nothing to evaluate. The architecture's three layers correspond to the three things this MCMC requires: the verification layer is the structural acceptance criterion, the institutional layer is the externalized representation of the smell that biases the chain (the atlas), and the pedagogical layer maintains the human capacity that produces the smell. The smell's function is biasing the proposal distribution toward productive regions; its form remains the trained judgment of the practitioner, beyond what an explicit prior could replace. The analogy is structural rather than technical. The Linter is a hard accept-or-reject filter, not a Metropolis acceptance ratio, and the architecture constructs no stationary distribution the acceptance criterion is designed to equal, so detailed balance and mixing times do not transfer; the claim is only that the two-component acceptance biases the proposal stream toward productive regions. Naming the smell as a prior on that stream also names a cost: a strong prior explores less, so the atlas can entrench the field's current sense of which regions are productive and make genuinely novel structure harder to propose. The smell's marking of sparse regions as worth exploring is a partial counterweight, a calibration choice rather than a structural guarantee, and the Shore is the mechanism that can demote an entrenched prior. The smell is load-bearing throughout the architecture, integral to its operation.

This is the response to the deeper version of Klowden and Tao's concern. At the individual level, AI-generated proofs risk lacking the narrative that lets a human reader understand them; the documentation onus addresses this. At the field level, AI-generated mathematics risks adding theorems without adding structural knowledge; the atlas addresses this. An AI-generated proof, admitted under the onus, contributes to the atlas regardless of whether any individual human has yet read it closely: its dependencies, technique clusters, and bridging structure enter the field's self-knowledge immediately. The field learns what it can do from the admission alone, and individual understanding follows through the atlas-surfaced connections.

The atlas also supplies the positive complement to the Compost Pile's negative records. Popper's World 3 is characterized in his account by the accumulation of knowledge through elimination; the Compost Pile operationalizes this negative accumulation. Mathematical knowledge also accumulates positively, through the structural enrichment of what has been established. The theory atlas is the institutional operationalization of this positive accumulation.

The atlas's role as shared institutional infrastructure across disciplines, and what that role implies for mathematics' position at this moment, is taken up in §6 after the architecture's internal consequences have been worked out.

4 Three Commitments for a Zero-Trust Verification Layer

4.1 Generators without Epistemic Agency

The verification layer must specify what it is protecting against, and the specification should be functional rather than tied to any particular account of how generators work internally. What matters is the absence of a verdict-to-weights channel during admission: the generator's parameters receive no update from the verifier's verdict while a claim is under consideration for entry into the verified record. This is the durable condition. The familiar picture of a large language model as a stateless flicker, each inference computationally atomic with no state persisting between events, is an approximation of current inference-time behavior rather than a permanent feature of the technology, and an agentic generator with long context, tool use, or multi-turn reasoning still satisfies the condition as long as its learned parameters hold stable against verifier verdicts during admission. What current generators share, under this approximation, is an optimization objective shaped by Reinforcement Learning from Human Feedback: the maximization of a reward signal correlated with human approval. In mathematics, human approval correlates with the surface aesthetics of argumentation: clear exposition, logical-seeming flow, confident conclusions, the syntactic markers of rigor.

This optimization creates a systematic incentive for odorless output of both kinds identified in Section 1. When the generator encounters a structural gap, its reward-maximizing strategy is to bridge the gap with language mimicking valid reasoning. This produces odorless proofs of the first kind: corked bottles with attractive labels. The mechanism is specification gaming (Krakovna et al., 2020): when the true objective (valid proof) is intractable and the proxy (text resembling a valid proof) is tractable, the optimizer converges on the proxy. Even when the generator produces a genuinely correct proof, its optimization toward surface aesthetics provides no incentive to produce the causal narrative, the structural insight, the honest engagement with difficulty that distinguishes a mathematically *interesting* proof from a merely *correct* one. This produces odorless proofs of the second kind: sound wines with no character.

The architecture is therefore robust to statefulness in the ordinary engineering sense, and the exposure is narrow and specific. It opens only if the training regime is restructured so that verifier rejections become a training signal in the admission loop itself, the one arrangement that would open a verdict-to-weights channel current deployment leaves closed. That restructuring, not statefulness as such, is what the verification layer must watch.

A repository's competitive position in the market for queries depends on the reliability of its admissions. A repository that admits gamed proofs loses reputation and query value; a repository that detects and flags them gains both. The filter evolves in response to attempted gaming as the repository's rational response to its own incentives, analyzed in §5.2.

4.2 Commitment 1: Verification Asymmetry

The first commitment formalizes the asymmetry between synthesis and verification. Generating a novel proof is computationally hard; verifying a fully specified proof term is tractable. This asymmetry is the architectural foundation of the zero-trust boundary. The generator proposes candidate proof terms. The proof-assistant kernel renders the verification verdict.

The restriction is enforced mechanically. Because the generator cannot learn from consequences during inference, behavioral governance is structurally inapplicable. Governance operates through the geometry of the output space: the verification boundary is a topological constraint on the set of outputs that can enter the group's verified record.

The proof-assistant kernel (Lean 4, Coq, Agda) provides the required anchor. The compiler checks the proof term against the type signature. It checks whether the proof compiles. This is Mayo's (2018) severity at maximum strength: the test's capability of detecting flaws is total for any structural inconsistency expressible in the type theory.

The computational asymmetry has an institutional counterpart: the onus of formalization sits with the generator (§3). The inversion is what makes the institutional layer (§5) tractable: once the verifier's cost is cheap, the function can be absorbed at scale by repositories, and journals are freed to specialize in curation. The onus is discharged modularly. The generator formalizes their contribution conditional on honestly declared axioms, which makes the architecture deployable without waiting for full library formalization.

4.3 Commitment 2: Formal Grounding

The second commitment enforces grounding: every claim entering the verified record must correspond to a formal statement for which a proof term type-checks against the trellis's formalized typeclasses. A claim existing only in natural language with no corresponding formal proof term is a floating semantic token.

The formalization gap, the boundary between what the type theory can and cannot express, is an honest limitation. The gap is documented in the Compost Pile as a meta-level entry, recording which constraints have been formalized, which resist formalization, and what the obstacles are. Claims outside the compiler's jurisdiction are audited with reduced epistemic authority, and this reduction is made explicit.

4.4 Commitment 3: Recursive Self-Assessment

A system that audits hypotheses against a fixed trellis, without periodically auditing the trellis itself, accumulates dead wood: constraints that once served the vineyard's conditions but no longer reflect the terrain. The compiler cannot distinguish convention from necessity; it enforces whatever trellis it is given. Only the Shore can perform that discrimination.

The Shore's signals are quantitative, drawn from the Compost Pile's metadata. Which constraints block the most hypotheses? Which blocks are class-level? Which reactivation conditions recur? The Shore also serves as a continuous adaptive defense against generators that might attempt to exploit the current trellis: if a generator adapts to the current constraints, the trellis shifts at the Shore's period, with axiom discharges and Class Block revisions changing the target between adaptation cycles. Upward correction from users who attempt formalization against the atlas supplies a continuous signal, so the target shifts not only at scheduled Shore events but whenever downstream use reveals that a declared axiom, a recorded dependency, or a chart-level claim was misstated.

4.5 The Cooperative Equilibrium and the Game 2 Objection

The three commitments jointly specify a complete zero-trust verification architecture. The equilibrium they engineer is a Game 1 equilibrium, over truth-seeking: generator and verifier each act within roles whose contracts govern the admission of claims to the verified library; the trellis and Compost Pile persist negative knowledge; the atlas provides the semantic-layer topology through which users navigate the library; modular axioms enable incremental deployment. The equilibrium is cooperative: no participant, human or algorithmic, improves their outcome by unilateral defection within the admission protocol.

The obvious objection is that Game 1 is not the game most mathematicians are playing most of the time. Mathematicians face career incentives in Game 2: hiring committees, funding bodies, journal prestige, citation economies. If the onus of formalization is a genuine cost borne before admission, why would any mathematician voluntarily accept it when Game 2 incentives reward publishing faster in venues that do not require formalization?

The response sits elsewhere than Game 2 resolution. Game 1 provides a distinct and accumulating form of value available to any mathematician who pays the onus, independent of whether Game 2 has realigned around that value. A proof admitted under the onus enters the theory atlas, which means its dependencies, its position in technique clusters, its bridging relations, and its axiom dependencies

become queryable by every subsequent user of the repository. This is a new kind of contribution that the informal-journal track does not produce: a contribution to the field's self-knowledge about what mathematics can do, institutionally legible and compounding over time. The onus is paid; the atlas is received.

The formalization effort is already underway. Mathlib, Isabelle's Archive of Formal Proofs, Coq's standard library, and Mizar's Mathematical Library are community-maintained verified libraries that grew through volunteer contribution without realignment of hiring, tenure, or grant allocation. Individual mathematicians are paying the onus of formalization now, for reasons the current academic reward system does not explain. The architecture's adoption question is therefore whether institutions participate in infrastructure being built regardless of their participation. Different institutional actors face different self-interest calculations: departments face credibility pressure as AI-generated content enters the literature; funding bodies face accountability pressure to demonstrate reliability of funded research; journals face editorial capacity pressure as peer review is outpaced by generation. The architecture responds to each of these pressures. Mathematics as a discipline faces a related pressure with particular weight: its standing with the sciences that depend on its patterns requires that mathematical content remain trustworthy at scale, which is what the architecture secures. The individual researcher's Game 2 calculation follows the institutional environment the champions create.

Whether Game 2 institutions eventually realign their incentive structures is a free-tier question by the partition of Section 1, and the architecture holds no position on it. Competitive dynamics among repositories (Section 5) provide the mechanism by which demonstrated atlas quality becomes a reputational asset institutions have reason to attend to. Funding bodies already attentive to reproducibility and to open-science commitments are natural early movers; the repository offers them a verification standard they can reference. The architecture's primary output holds independent of any realignment, which is the point of putting realignment in the free tier: the value accumulates in the certified tier, and what the field does with it is the field's to decide.

5 Consequences: Repositories, Journals, and Pedagogy

5.1 Repositories as Verifiers, Cartographers, Coordinators

Once verification is mechanized under the onus principle, three institutional consequences follow from what a repository holds and how it holds it.

Verification hygiene. The repository confirms that admitted proof terms type-check and that the axiomatic base is auditable. A generator cannot silently add axioms, weaken definitions, or stretch typeclasses to get a term admitted. Every dependency is visible in the compiled artifact, and declared axioms are themselves formal objects tracked with provenance, scope, and discharge status. Under the

onus principle, verification is the admission criterion; its severity is total for any inconsistency expressible in the type theory.

Cartography. From the verified library, the repository constructs the theory atlas. The atlas is a research tool in a sense the library itself is not: the library records individual results; the atlas records their structural relationships across multiple consistent charts, making the field's self-knowledge about its own productive capacity queryable. At AI-generation rates, the atlas is also what makes curation tractable. Structural deduplication, where proofs that differ syntactically but are isomorphic under a chart's compatibility relations can be grouped as clones of an abstract pattern with departure metadata, reduces the pragmatic layer's input from "every verified proof" to "every structurally distinct family of verified proofs." Functional-specification retrieval lets curators ask the atlas for proofs satisfying a particular typed need, collapsing open-ended search into targeted query.

Coordination. Declared axioms across the submitted library constitute a demand signal for the volunteer formalization effort. Heavily cited undischarged axioms rise in priority; rarely cited ones can wait. This gives the existing Lean/Mathlib community and analogous efforts a coordination mechanism they currently lack: formalization priorities emerge from demonstrated research demand rather than from the individual judgment of contributors working in parallel. The repository does not manage the formalization effort; it surfaces which undischarged axioms are load-bearing across the verified library.

5.2 Red-Teaming as Competitive Function

Klowden and Tao (2026) distinguish a "blue team" that generates content from a "red team" that verifies and probes for weaknesses, and argue that AI is safer in red-team roles than in blue-team ones. The architecture operationalizes this distinction at the level of proof-term admission. The proposer is the blue team; the repository's type-checker and red-teamers are the red team, testing whether the proof compiles, whether declared axioms are accurately stated, whether documentation is adequate, whether the proof's atlas position is coherent with its claimed dependencies.

A repository can accept passively (the axioms compile, documentation is present, admitted) or actively (the repository's own scrutiny tries to find ways the declared axioms are misstated, scoped incorrectly, or more load-bearing than the proposer indicated). Active red-teaming produces a higher-quality atlas. A repository that accepts passively and later has to retract when discharge reveals axiom errors suffers a cost in atlas quality that a more rigorous competitor does not. Over time, repositories develop reputations for how often their admissions survive downstream discharge cleanly, and the value of a query against a repository is a function of that reputation. This is the market mechanism by which the red-teaming function is self-sustaining and by which atlas quality becomes a reputational asset.

5.3 A Sorry on the Repository's Governance

The architecture's most significant unresolved obligation lies at the boundary between Game 1 and Game 2: the governance of the repositories that build the atlas. Two structurally distinct problems hide under one heading, and they carry different sorry types.

The first is steady-state governance: capture, gatekeeping, and oligopoly among repository operators once several repositories already exist. This one resolves rather than defers, and it resolves into the free tier. Which repositories a field trusts, which atlases it consults, and what an operator's standing is worth are judgments the field makes and revises, and they recompute from nothing, so the architecture specifies no mechanism for them and specifying one would install the target the partition exists to avoid. What the architecture owes here is the conditions under which the judgment can be exercised, which are that atlas quality is publicly checkable, that a rival can be founded, and that content can leave. Ostrom's Design Principle 3, collective-choice arrangements authored by those the rules bind, is the empirical form of this: the governing judgment belongs to the practitioners the governance falls on, and the architecture's task is to keep that judgment available rather than to make it for them.

The second is bootstrapping, and it is a gate sorry rather than a deferred one. Who builds the first atlas, and what prevents capture during formation, before competitive dynamics have any rivals to work on. Atlas value grows with coverage, which is a winner-take-all dynamic, so network effects in atlas construction can produce oligopoly faster than competition can correct it. The §5.2 argument is silent here by construction, because it presupposes the plurality that bootstrapping has not yet produced. This sorry blocks the deployment claim for the cartography layer, and it should be read as gating rather than deferred.

The gate is bounded, and the boundary is worth stating precisely, because it falls on the cartography layer and not on the verification layer. The onus and the modular-axiom discipline deploy without any atlas at all: a single author formalizing a contribution against honestly declared axioms produces a verifiable record that any party can recheck by rerunning the kernel, and that verification is capture-resistant because no repository owns the verdict. What bootstrapping threatens is who controls the shared structural map, and the institutional-lead claim that rests on it, not whether the proofs check. The verification layer survives repository capture; the cartography layer is what the gate sorry holds open.

What the crossing rule of §3.4 supplies is a bound rather than a discharge, and the bound is worth stating in the form it actually holds. A captured atlas exports results that ground nothing elsewhere until a local author retypes them under local constraints and signs the retyping, so capture stays contained to the repository that suffers it and remains visible in the record, forkable by anyone who rejects it, and costly to whoever attempts it. Capture itself remains possible. Resistance to it is a property maintained continuously by distributed authorship and open checking rather than a guarantee the architecture issues once. The gate sorry is therefore narrower than it was: it holds open who authors the first atlas and what disciplines that authorship during formation, and it no longer holds open how far a capture propagates.

5.4 Dependency Chains and Citogenesis

Klowden and Tao describe citogenesis in mathematics as a recursive contamination of the evidentiary base: their case is that deep-research tools used to catalog open problems began citing those catalogs back as authoritative, with the effect that updates to the catalogs interfered with subsequent literature searches. The axiom-dependency chart addresses the misstated-axiom case directly: when a declared axiom is discharged and found to have been misstated, the chart identifies every result that depended on the axiom, directly or transitively, and flags each for revision. Current mathematical practice treats a retraction as a social event that hopes to reach downstream authors; the architecture treats it as a traceable dependency update.

The recursive-contamination case is addressed by a different feature of the architecture. The repository's admission criterion is orthogonal to the informal discussion layer where citogenesis operates. What the library contains are type-checking proof terms under declared axioms; what the atlas records are the structural relationships among them. Informal citations, deep-research summaries, and natural-language claims do not enter the library or the atlas through admission. The recursive contamination Klowden and Tao observed is a consequence of allowing informal secondary summaries to function as primary references. Under the architecture, no informal summary achieves primary-reference status; only the type-checking proof term does. The evidentiary base is, by construction, uncontaminated by its own secondary descriptions.

This is the structural form of a problem now visible in practice. Preprint repositories facing AI-generated submissions have begun policing fabricated and hallucinated citations after the fact, through author sanctions for negligent references. A declared axiom under the architecture cannot be a hallucinated citation, because it carries provenance, scope, and a discharge status, and a fabricated source fails at the first discharge attempt rather than propagating silently. The sanction taxes the symptom after admission; the architecture closes the failure mode at admission. The two are complementary, but only the second addresses the cause.

5.5 The Reference Atlas

The typed-attestation discipline modeled in Appendix B is robust as a voluntary author practice, but mandating it at a repository inverts the citation-cost gradient. Where the current incentive over-cites, because citations are credit, a per-reference attestation cost incentivizes the reverse, and the references most likely to be dropped are the adversarial ones: the prior work that anticipates a result, and the work that conflicts with it, since the cheapest way to avoid writing a hard conflict-resolution is to omit the conflicting source. A discipline that asks authors to declare their conflicts, applied under a cost, rewards hiding them. The diagnosis that unifies this with the citogenesis case is a single quantity, the delta between the citation graph an author declares and the one the corpus actually exhibits. A hallucinated citation is a declared edge with no computed support; a strategic omission is a computed edge with no declared counterpart. Both are signs of the same mismatch.

The reference atlas is the repository's second cartographic output, the atlas principle applied to the corpus rather than to the verified library. Where the theory atlas maps the verified library, the reference atlas maps how the corpus attributes: each submission is placed in its neighborhood by typed edges, depends-on, extends, anticipates, conflicts-with, duplicates, computed from content rather than declared by the author. The author's reference list becomes an annotation on the computed graph rather than the graph itself, which is the forcing function the mandated discipline needs. An author can mark a computed edge irrelevant, but cannot shrink the visible neighborhood by shrinking the declared list, so omission becomes conspicuous rather than free. The apparent-conflicts section in particular stops being author-curated and becomes repository-proposed: the atlas hands the author the candidate anticipations and conflicts from the computed neighborhood, and the author must resolve or rebut each. The cost that drove minimization collapses in the same move, because the atlas has already computed the neighborhood and can draft the typed attestation for review rather than production.

One thing is weaker here than in the theory atlas and must be stated with the construct. The theory atlas maps a formal substrate with verified edges; the reference atlas maps a social substrate with estimated edges, since computing anticipation and conflict from natural-language claims is fallible. The reference atlas is therefore a proposal layer rather than a verification layer, the proposal distribution rather than the Linter, and its edges are contestable by construction.

The second thing is not a weakness under the partition of Section 1, and recognizing that is what the construct is for. The reference atlas reaches into Game 2, because references are credit and the anticipation chart is where priority is settled. Priority is the Game 2 element that recomputes: admission is dated, proof terms are content-addressed, and the anticipation neighborhood is computed from content rather than declared by an author. So this is the one place the architecture certifies a credit question, and it certifies it for the same reason it certifies anything, which is that a recomputable mechanism removes the position worth capturing. Whoever controls a declared citation list controls priority; nobody controls a computed neighborhood that any party can rerun. The gradient the construct sits on is real, running from a formal substrate to a social one and from verified edges to estimated ones, and it is why the reference atlas proposes rather than adjudicates. What it does not do is carry the architecture from truth into credit, because priority was already on the certified side of the partition and the atlas is how it gets computed.

5.6 Coupling, Generativity, Fragility

Documentation conventions populate the atlas's technique clusters. Atlas construction produces the isomorphism hypotheses red-teaming tests and the retrieval structure proposers use. Repository red-teaming produces the reputational signal proposers weigh and the verified content the atlas aggregates. Each practice pays forward into multiple others, which is the source of the architecture's stability: a practice under competitive pressure is defended on its own merits and by the value it produces for practices that reinforce it in return.

The coupling is generative. Better documentation yields a richer atlas. A richer atlas surfaces more isomorphisms and bridging structure. Surfaced bridges let mathematicians transport techniques from one region of the library to another, solving problems the original techniques were never applied to. Those solutions enter the library with their own documentation, at a baseline the previous cycle raised. Over time, the field accumulates not only more verified results but progressively more legible structural knowledge: techniques cluster more tightly, bridges become more numerous, the landscape becomes easier to navigate. This is what the architecture produces that mechanical verification alone cannot: a ratchet on the structural understanding of the field.

The coupling is also the source of the architecture's fragility. If adherence to any one practice falls below a threshold, the practices it supports lose value and their own support diminishes, and the arrangement can unravel through the same couplings that sustained it. The architecture is robust under adherence, generative under compounding, and fragile under neglect.

Coupling is an interdependence claim, and the architecture rests on a stronger one, which is that each of the three layers is load-bearing. Take each away in turn. Without the onus, the verifier's cost stays unbounded and admission reverts to what bounded reviewer effort can check, which is the arrangement whose failure under AI generation the paper opened by diagnosing. Without the atlas, admission still works and the field accumulates verified theorems while accumulating no structural knowledge about what it can do, which is the field-level form of Klowden and Tao's concern and the one mechanical verification alone leaves untouched. Without the pedagogy, no operator can maintain the Shore, evaluate whether a declared axiom is faithfully stated, or read what the atlas returns, and the two remaining layers run without anyone competent to revise them. Each removal breaks a different thing, which is what distinguishes a minimal architecture from a set of practices that happen to reinforce one another. The three layers are the least that closes the variety gap, and the claim is refuted by exhibiting an arrangement that closes it with fewer.

5.7 The Repository's Product

The three functions converge on a single deliverable. The repository's product is not its archive of verified proofs, though it holds that; not its ledger of conditional eliminations, though it maintains that; not its index of declared axioms and their discharge status, though it tracks that. The repository's product is the positive topology of mathematical knowledge: the structural relationships among verified results, the technique clusters and bridging patterns, the dependency structure and the axiom-dependency structure, the cartography of the library as a navigable landscape. The product compounds: each admission adds to the topology, each newly surfaced bridge enables solutions that add further admissions, and each cycle begins from the landscape the previous cycle enriched. The repository operates a ratchet on the structural knowledge of the field.

5.8 Journals in Curation's Purified Form

The separation of verification from curation is the budget of Section 1 applied at the institutional scale where the scarce resource is actually spent. Editorial judgment is the field's rival good: an editor's attention consumed on whether an argument holds is unavailable for whether it matters, and the two are not substitutes. Mechanizing the first is what makes the second affordable. The gain does not come from editors becoming better curators; it comes from the certifiable work leaving their budget. Ostrom's Design Principle 4, monitoring accountable to the users whose resource is being monitored, names what makes the arrangement stable: the field checks the repository's admission record and the record is open, so the appropriation of the field's judgment is itself visible to the field.

Relieved of verification burden, journals specialize in curation. They select from the verified library (and its conditionally verified content) which correct results warrant the field's attention. Journal quality is measured by curatorial judgment, not by verification rigor. Under mechanized admission, the AI-or-human provenance of a proof is not a criterion for admission: the proof either type-checks under declared axioms with documentation meeting the repository's standard, or it does not. Journals may, as a curatorial matter, prefer to highlight human-authored work or require disclosure of AI involvement; that is a pragmatic-layer decision about what gets attention, and the paper does not prescribe it. The verification layer does not discriminate by author.

Current journal practice confounds verification and curation. An editor assessing a submission's importance is simultaneously assessing its structural correctness, because no other institution has mechanized the correctness function. The confounding has consequences: a structurally correct but unimportant paper can be rejected on the grounds of importance, and a structurally flawed but important-seeming paper can be accepted on the grounds of promise, with reviewers expected to catch flaws under bounded effort. The resulting literature carries both false negatives and false positives. Under the separation, the two functions operate independently. The repository admits; the journal selects. Curatorial judgment is what the journal offers, and that judgment can be evaluated on curatorial terms.

The purified form has its own failure modes: an editorial culture can ossify, curatorial judgment can reflect in-group bias, a field can allocate attention poorly even when every selected paper is structurally sound. These are curatorial pathologies the architecture leaves to curatorial governance. What the architecture does is stop making them worse by lifting the verification burden that currently distorts editorial capacity.

The separation between verification and curation is analytical rather than complete. The two layers couple through a feedback loop the architecture cannot eliminate: the verified library's content shapes the pool from which curators select, and the standards of interest that curators apply shift as the pool's character shifts. AI-generated proofs entering the verified library at high volume change the texture of what curators face, and the criteria of interest will adjust in response: toward proofs that connect previously distant atlas regions, toward proofs whose technique-cluster position is novel, toward proofs whose departure-compost-pile interactions reveal previously hidden phenomenological gaps. The atlas itself becomes a curatorial tool, and its metrics become proxies for interest in ways the paper does not

fully specify. The separation of functions remains useful as an analytical distinction; the institutional reality is coupled, and the coupling is part of what makes the architecture an evolving rather than static arrangement.

5.9 Pedagogical Consequences

The architecture presupposes human operators capable of maintaining the Shore, interpreting the atlas, and exercising judgment about axiom declarations. Those operators are produced by a pedagogical process that must include direct encounter with failure, struggle with unmapped terrain, and the discipline of declaring honest dependencies. Whether labor-market incentives allow this pedagogy to be sustained at institutional scale is a Game 2 question; the paper specifies what pedagogy the architecture requires and leaves the political economy of its maintenance to the infrastructure that handles Game 2.

The risk takes three forms. **Compost Pile dependence**: students who always query before attempting never develop the pattern recognition that makes the Compost Pile a research tool rather than a crutch. **Atlas dependence**: students who always navigate by the atlas never develop the structural intuition that lets experienced mathematicians see connections the atlas has not yet recorded, and the atlas will fail to grow. **Axiom-declaration abdication**: students who always lean on conditional formalization never develop the judgment required to know when an axiom statement is well-formed, faithful to its referent, and scoped appropriately.

The atlas is also a pedagogical medium, and a genuinely different one than mathematics has had before. Current curricula are organized historically and subdisciplinarily: calculus, linear algebra, real analysis, abstract algebra, topology, each taught as a self-contained body. The atlas exposes what these historical taxonomies leave tacit: the structural patterns techniques instantiate across subfields, the bridges connecting nominally separate areas, the functional specifications that organize productive capacity. A student who queries the atlas for techniques that prove uniqueness results gets a cross-subfield cluster currently reached only through years of cumulative intuition. The atlas is valuable as a medium of exposure to the field’s structural self-knowledge; it supplements rather than replaces the work of learning. A student who queries without engaging the underlying mathematics is a tourist reading guidebooks without traveling.

The trellising discussed so far has been organized for the pursuit of knowledge creation, but one could also organize the trellis for pedagogy. A pedagogical trellis is a training environment engineered to expose a particular anti-pattern or teach a particular structural lesson, with constraints chosen to target the capacities being trained: one trellis to develop the discipline of honest axiom declaration, another to develop structural intuition without atlas dependence, another to calibrate the student’s own Compost Pile before they query the institutional one. The student proposes hypotheses, encounters mechanical blocks, extracts the residue that survived, tries again. The encounter with the block is the lesson. This is closer to case-based pedagogy in law and medicine than to current mathematical pedagogy, and it provides precisely the Compost Pile-free, atlas-free, and axiom-declaration-free contexts the triple risk

above requires, without abandoning the architectural discipline entirely. Pedagogical trellises are proposed here as the natural extension of the research trellis into training contexts; their development would be a research program in mathematical pedagogy that the architecture makes tractable.

The smell does not vanish under the architecture; it specializes upward. Historically, the trained mathematician's smell performed first-order fault detection on individual proofs: is this argument structurally sound, does this step do the work it claims to do, where might this proof have hidden a flaw. Mechanized verification absorbs first-order fault detection. What remains for the trained nose is higher-order judgment: which type-checking proofs warrant deep engagement, where the atlas's bridges connect formally without illuminating substantively, which technique clusters mask categorical differences the chart has not yet recorded, where the departure compost piles are likely to find traction. This is a different smell than the historical one, calibrated to a different question. The pedagogical layer's task is to produce mathematicians whose noses are trained for the higher-order judgment the architecture cannot mechanize. The Compost Pile-free, atlas-free, and axiom-declaration-free contexts the pedagogy requires are the training environments where this specialized smell develops; the pedagogical trellises of the previous paragraph are the structured form those environments take.

Honest dependency declaration is a form of epistemic humility made mechanical. A student who has internalized the discipline of declaring what they actually assume, rather than gesturing at a citation and moving on, has internalized something structural about how mathematical knowledge accumulates. The pedagogical layer transmits the discipline the architecture embodies, beyond preserving expertise. The architecture's reach extends beyond the mathematical classroom. The atlas is shared infrastructure across disciplines; the pedagogical implications extend to students in any empirical field whose subject matter engages mathematical patterns. The larger relationship between mathematics and the other sciences under the architecture is the subject of §6.

6 The Sommelier of Knowledge

6.1 Recapitulation

Klowden and Tao diagnose AI as threatening the culture of mathematical understanding. The architecture specified in this paper responds with three layers resting on one principle. The principle is the onus of formalization: whoever wants a claim admitted to the verified record produces a type-checking proof term, accompanied by documentation meeting the repository's standard, conditional on honestly declared axioms. The verification layer mechanizes what used to be the reviewer's burden. The institutional layer reorganizes repositories around the verification function they now absorb at scale, produces the theory atlas as their characteristic output, and frees journals to specialize in curation. The pedagogical layer preserves the human capacity the architecture presupposes. These are the internal consequences: the response to AI as it affects mathematics itself. The external consequences extend further.

6.2 Mathematics as Certified Compression

The institutional-lead claim has so far rested on mathematics' subject matter being structural patterns abstracted from substrate. That is true and weaker than what is available. Stated as a characterization the architecture holds independently of, and one formalists will grant more readily than structuralists or platonists: mathematics is the lossless compression of the infinite, and a proof is the certificate of losslessness.

A theorem is a finite encoding of an infinite family of instances. The quantifier prefix specifies the family, the statement supplies what holds of each member, and the proof certifies that every member is recoverable from the encoding. Losslessness is relative to a declared reconstruction target in mathematics as in every other use of the word, and the theorem statement is where mathematics declares its target. An existence theorem targets the proposition and recovers it exactly; a result on asymptotic behavior targets the limiting ratio and recovers it exactly. Independence results are the record of where compression is unavailable rather than where it is degraded, which is why they sit in the Compost Pile as entries at the highest tier and fill its schema without strain: the blocked hypothesis is the statement, the blocking constraint is the axiom system, the mechanism is the independence proof, the block scope is class rather than point, the residue is the technique together with the relative consistency result, the reactivation condition is a strengthening of the axioms, and the trellis version is the axiom system itself.

Two consequences carry this section. The first is that mathematics is the limiting case of one operation rather than a different kind of activity. Section 1 defined knowledge as what survives attenuation of the producer, and a theorem is where that attenuation runs to completion, because the certificate regenerates the entire family with no reference to who wrote it. Universal quantification performs the same operation along a second axis, attenuating the particular instance to the point where what remains holds of every member. The empirical disciplines run the operation and stop short, since the world keeps supplying instances and signs off on none of them. Mathematics holds the shared structural layer because it is where the shared operation reaches its limit, which grounds the custodial role more firmly than substrate-abstraction does on its own.

The second is that the certificate is internal because mathematics answers to nothing outside itself. This is a claim about answerability rather than about existence, and it explains an asymmetry the previous sections observed without accounting for. A theorem states the domain over which its decoding is faithful, so an error is internal and detectable by the discipline's own apparatus. A physical law's domain of application is learned from the world and always after the fact: Newtonian mechanics remains exactly as true as it ever was considered as mathematics, and what failed was the binding of that structure to measured quantities beyond a boundary nobody had specified in advance. Mathematics certifies its own boundary; the empirical disciplines discover theirs. This is also why a compression with no referent can be certified from inside, since everything the certificate quantifies over is present to it, and why a compression with a referent cannot.

The traffic between mathematics and the empirical disciplines becomes legible in the same vocabulary, and it runs in both directions. A working compression arriving from physics without a certificate is a declared axiom in this architecture's exact sense: a placeholder carrying provenance, in use, flagged undischarged, and constituting demand on the formalization effort. The delta function was such a placeholder for decades before distribution theory discharged it. Archimedes drew the line himself in The Method, obtaining the volume of the sphere by mechanical balancing, recording that the mechanical route furnishes no demonstration, and supplying the geometric proof separately. The operation carries the label rather than the practitioner: a physicist who supplies a certificate is doing mathematics in that moment, and a mathematician conjecturing is doing what the empirical disciplines do.

6.3 The Atlas as Shared Infrastructure

The empirical sciences engage phenomena that instantiate functional patterns, and mathematics is where those patterns are specified most cleanly. A biologist modeling yeast population dynamics, a materials scientist modeling heat dissipation in a bounded substrate, and an epidemiologist modeling contagion through a network are engaging different instantiations of structural patterns the atlas addresses. Currently, each discipline accesses mathematical technique through its own applied-mathematics subculture, maintained at significant cost and with uneven coverage. The atlas offers a public institutional alternative: a researcher states the structural features of their problem and receives the technique cluster mathematics has developed for that pattern. The claim here is structural, not reductive: each discipline retains its own domain vocabulary, theory, and methodology, and the atlas supplies the shared structural layer these disciplines already engage. The patterns shared across disciplines, long recognized as the ground of mathematics' "unreasonable effectiveness" (Wigner, 1960), have an institutional form in the atlas that makes the sharing queryable.

6.4 The Grounding Moment

The institutional-lead claim needs its mechanism stated, or it reads as assertion. The mechanism is a pairing invoked at a specific moment. Each empirical discipline maintains its own trellis, configured for its objectivity level: empirical rather than purely formal commitments, Linters that flag rather than reject, verification that remains partly judgment-dependent. The mathematician is not embedded in that trellis and does not adjudicate the discipline's own claims. The pairing is invoked when the domain specialist needs to ground their structural map to the territory, when they need to know whether what they are calling a pattern is in fact a mathematical structure, or whether a model's failure is a phenomenological departure from the world rather than an error in the formalization. The atlas answers the first question by returning the structure, if there is one, and its known properties. The grounding event resolves the second by separating departure from mistake.

The departure compost piles of the next section are the output of these grounding events. A grounding event that ends in departure rather than error produces a record in the domain's native vocabulary, which

the pairing translates into a mathematical demand signal. Stated this way the institutional-lead claim follows rather than asserts: mathematics is not absorbing biology's verification crisis, it is maintaining the shared structural layer biology already uses informally, making it queryable, and providing the interface through which departure knowledge flows back as demand on mathematical development. The lead is custodial rather than jurisdictional. Mathematics holds the map; each discipline keeps authority over its own territory.

The mechanism surfaces an open dependency the paper has not discharged. The grounding moment requires a practitioner who can read both the domain trellis and the atlas: a domain specialist with enough mathematical fluency to pose a grounding question precisely, or a mathematician with enough domain knowledge to translate the atlas's answer back into the discipline's vocabulary. The pedagogical layer of Section 5 trains the mathematician who maintains and interprets the atlas. It does not train this bilingual practitioner, whose competence sits across two trellises rather than inside either. This is a second-order pedagogical problem, distinct from both the domain trellis and the atlas, and the paper declares it as a sorry: the interface the institutional-lead claim depends on has no specified pedagogy, and the claim is conditional on that pedagogy being constructible.

6.5 The Departure Compost Piles

The relationship is bidirectional. The applied disciplines produce a distinctive form of knowledge the pure-mathematics community is not positioned to produce: where specific phenomena depart from the patterns the atlas returns. The biologist knows where logistic growth fails as a description of yeast dynamics, under what conditions the failure matters, and what features of the phenomenon the mathematical pattern does not yet capture. This knowledge is structurally analogous to the Compost Pile but directed at a different object: the Compost Pile records where hypotheses fail structural constraints; a departure compost pile records where structural patterns fail phenomenological reality. Each applied discipline could maintain its own departure compost pile at the interface with the atlas, recording where phenomena leave its techniques behind, what residue the departure leaves, and what mathematics would need to do to capture the phenomenon more faithfully. These records would serve both directions: they would flag for applied users where existing mathematical treatment is known to break down in their domain, and they would constitute structured demand signals for mathematical development, specifying precisely what mathematics does not yet do. A departure compost pile has a complement that flows in the same direction and carries the opposite content. Where the departure record holds cases in which a certified compression fails against phenomena, the complement holds cases in which a working compression lacks its certificate: the manipulation rules established in practice, the technique that reliably produces correct answers, the structure a discipline uses daily and cannot yet ground. Both are demand signals on mathematical development, and the architecture already has the object for the second, since an uncertified compression in use is a declared axiom awaiting discharge. The interdisciplinary infrastructure is collaborative: the atlas provides portable structural patterns, and the departure compost piles provide the phenomenological knowledge of where those patterns leave specific realities behind.

As AI generation accelerates atlas additions, the departure compost piles become critical infrastructure for applied science, preventing uncritical adoption of AI-generated mathematical content by disciplines that should be checking where the content departs from their reality.

6.6 The Moment

The timing matters. AI generation is forcing every empirical discipline to confront the inadequacy of its current verification infrastructure. Under the requisite-variety framing, this is one event with one shape: AI is amplifying the variety of generated content beyond what the existing institutional regulators in each discipline can absorb. Mathematics has a specifiable institutional form that closes the variety gap and that could organize the shared structural layer across disciplines. No other discipline is positioned to play this role: the empirical sciences are committed to their specific substrates, statistics is a narrower toolkit specialized for inference under uncertainty, computer science provides the verification infrastructure but not the subject matter. Mathematics is distinctive because its subject matter is structural patterns abstracted from substrate, which is exactly what the interdisciplinary infrastructure role requires. Under the architecture, mathematics takes institutional lead by maintaining the shared structural layer the sciences already engage, while each discipline keeps its own theory, vocabulary, and methodology. Mathematics keeps the atlas and participates in the departure compost piles as the institution where the sciences' demands on structural knowledge become legible. This is an available posture for mathematics at this moment that it has not had before and may not have again in the same form, because AI is the forcing function that exposes every discipline's current arrangements as variety-deficient while the formalization community has quietly built exactly the apparatus the architecture requires.

6.7 Kuhn's Diagnosis Through Ashby

Kuhn documented two distinct things in his account of revolutionary change: the substance (rapid foundational reorganization, broad reframing of prior work, the discipline operating on different ground after the revision) and the experience (the felt collapse of foundations, the protracted period of incommensurability, the piecemeal rediscovery of load-bearing dependencies). The disciplines he studied bound the two together because their institutional apparatus lacked the variety to absorb foundational change without breakdown. Ashby's law of requisite variety specifies the principle: a regulator can govern a variable only when the regulator's variety meets or exceeds the variety of the variable being governed. Tacit foundations carry variety no informal institutional response can match, so a foundational revision under tacit conditions overwhelms the institution and produces the phenomenology Kuhn documented. Explicit axiom declaration with tracked provenance closes the requisite-variety gap. The institution's apparatus, the trellis plus the Compost Pile plus the axiom-dependency chart, has variety sufficient to absorb the substantive reorganization that historically overwhelmed institutional response. A high-tier Shore event reorganizes the field's foundations; the dependency chart specifies the scope of consequence; reformalization proceeds against the new

foundation as a tractable construction project. The architecture enables the conditions for Kuhnian revolution in mathematics while shedding the catastrophe Kuhn observed in disciplines without explicit axiom declaration. The substantive revolution is preserved; the institutional catastrophe is shed. Kuhn deliberately bracketed the question of whether mathematics historically exhibited revolutions in his sense; the architecture renders the historical question moot by specifying the institutional conditions under which revolutionary substance and revolutionary catastrophe become separable for any discipline that adopts them. Kuhn's documentation of the disciplines he studied is accurate. Those disciplines lacked the requisite-variety apparatus the architecture supplies. The architecture's contribution to the philosophy of scientific change is the demonstration that revolutionary substance and revolutionary catastrophe are separable when the institutional regulator has variety to match the change being regulated. Mathematics under the architecture has access to the substance without the catastrophe. Declared axioms capture the formal component of what Kuhn meant by the disciplinary matrix; exemplars, values, and broader disciplinary culture remain beyond what axiom declarations can formalize, and the cultural component continues to require human transmission through pedagogy. Within that bounded scope, the other sciences have the architecture available as a model for the formal component of their own institutional practice.

6.8 The Sommelier of Knowledge

This paper opened with a sommelier doing three things: catching structural faults in wines before they reach the guest, curating the list upstream of service, and recommending wines in dialogue with users who knew what they wanted but lacked the vocabulary of the cellar. The architecture specifies how mathematics performs all three at the level of disciplinary infrastructure. The closing move is that mathematics, under the architecture, becomes the sommelier of knowledge for the sciences more broadly: the institutional form that catches structural faults in its own proofs, curates what merits the field's attention, and recommends to researchers from other disciplines the structural patterns that fit the phenomena they engage.

The three functions land at different layers. Mechanized verification performs the custodial fault detection that the sommelier performs at the bar: the proof either compiles or it does not, the structural integrity is checked before admission. Journals, freed of the verification burden, perform the curatorial judgment the sommelier performs upstream: what merits the field's attention given its identity, its accumulated commitments, and its forward direction. The atlas, maintained by repositories and read by trained mathematicians, performs the recommendation function: a researcher arrives with a phenomenon, states its structural features, and receives the technique cluster mathematics has developed for that pattern, with the trained interpreter helping to read the atlas's response in light of the researcher's specific situation. The architecture distributes across institutions and across human and mechanical components what the sommelier performs as one trained capacity. The distribution is what makes the architecture work at the variety AI now generates; the integration of the three functions, recovered

through the institutional structure that holds the components together, is what makes mathematics a sommelier of knowledge rather than a collection of automated services.

The mapping is a guide rather than an identity. Two breaks deserve naming. First, the sommelier's recommendation is bidirectional in real time: she asks follow-up questions, watches the guest, suggests a tasting flight to calibrate. The atlas, even read by a trained mathematician, returns a technique cluster in response to a structural query without iterative dialogue at the same temporal scale or with the same affective sensitivity. The trained interpreter recovers some of the iteration; the architecture's recommendation infrastructure remains more static than the sommelier's service. Second, the sommelier holds all three functions in one trained perceptual capacity, and her coherence across functions, the way her recommendation is informed by her fault detection which is informed by her curation, follows from the integration in one perceiver. The architecture's integration is institutional rather than perceptual. Whether institutional integration suffices for what perceptual integration delivered is genuinely open; this paper claims that it does, for the substantive work mathematical practice requires, but the claim is empirical and the demonstration will come from the architecture's deployment rather than from the architectural specification alone.

This is the role available to mathematics at this moment. Whether mathematics takes it is a free-tier question, and the architecture leaves it there deliberately, because a discipline's judgment about what it should become is the last thing any mechanism should settle for it. Whether it can is the question the architecture answers. The answer is yes: the institutional form exists, the principle that organizes it is specifiable, the practices that sustain it are already in motion in the formalization community, and the moment at which the sciences need exactly this kind of shared infrastructure is now. This paper's contribution is the application of a foundational cybernetic principle to mathematical institutions: requisite variety, attenuated and amplified through the architecture, makes the substantive work of mathematical practice tractable at the variety AI now generates. The architecture's discipline throughout has been one rule stated at the outset and applied at every layer since: certify what recomputes from public structure, and leave free what does not, so that the field's judgment is spent only where nothing can be spent in its place. Verification is certified and interest is free. Priority is certified and prestige is free. The map is checkable and what a discipline makes of it is not. The same principle is available to the sciences whose verification arrangements will face analogous pressure. The sommelier is still standing. The table has become larger than she first surveyed.

AI Use Statement

This paper was written with AI systems, including large language models used for drafting, research, and source checking under the author's direction. The author conceived the argument, set its structure and commitments, made every substantive decision, and reviewed and revised the full text. The author takes full responsibility for the content, including any errors.

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8 Appendix A: Semantic Translation Table

The architecture introduces several terms that have partial counterparts in existing literature. The table below locates each construct in the traditions it extends, so readers from different backgrounds can orient quickly.

This paper's term	Closest existing analogs	Distinctive feature
Trellis	Lakatosian research program (hard core plus protective belt); typeclass hierarchy in dependent type theory	Revisable on evidentiary grounds; admits formalization as typeclasses
Compost Pile	Popper's World 3 operationalized; lessons-learned databases (NASA LLIS); file-drawer literature	Append-only, structured by blocking constraint, reactivation conditions, and trellis version
Shore	Kuhnian crisis; Beer's System 5 in the Viable System Model	Scheduled, evidence-driven; Compost Pile records survive with full metadata across revisions
Onus of formalization	Reviewer-side verification inverted; referee burden migration	Admission criterion rather than editorial judgment
Modular axioms	Conditional proof; assumption-parametric formalization	Each declared axiom carries provenance, scope, status; surfaced as demand signal for formalization community
Theory atlas	Mathlib typeclass hierarchy; Isabelle theory graph; dependency graph	Collection of compatible charts; positive counterpart to Popper's negative accumulation
Two Games	Scientific epistemic practice vs. scientific career economy (Merton-adjacent)	Architecture equilibriates Game 1 and partitions Game 2: priority certified, attention and reward free
Stateless flicker (verdict-to-weights channel absent)	LLM-as-oracle; computational agent without persistent learning during inference	Verification is robust to training statefulness but exposed to verdict-to-weights channels during admission
Reference atlas	Citation network; co-citation analysis; bibliographic coupling	Computed from content rather than author-declared; unifies hallucinated and omitted citations as the declared-versus-computed delta; a proposal layer, not verified like the theory atlas

The table is not exhaustive. It is offered as a key for readers familiar with any one tradition to locate the paper's constructs relative to that tradition without having to read the full paper first.

9 Appendix B: Verification and Provenance

This appendix applies the paper's own onus principle to the paper's own bibliography. The verification burden sits with the author rather than the reader: each reference is typed by its epistemic role, the specific use is stated, references that appear to contradict the paper's conclusions are listed with their resolutions, and fresh sources carry a status watch. The targets are the citation-graph failure modes the architecture is meant to make rare: bundling distinct sources under one marker, citing a review as a primary result, hardening a conjecture by attaching it to a source that does not contain it, citation cycles among concurrently circulated work, and references with no traceable use. Each entry below can be

checked mechanically against the body and against the cited source, and so the discipline scales where reader-side spot-checking does not.

The discipline is not new, and the appendix says so rather than presenting it as an invention. Moravcsik and Murugesan (1975) classified citations by function fifty years ago, and CiTO (Shotton, 2010) implements typed citation in a machine-readable ontology today. What this appendix adds is the author-side obligation: the typing is produced by whoever wants the admission rather than reconstructed afterward by a bibliometrician, and the apparent-conflicts section makes the author state and resolve the adversarial neighborhood rather than leaving it to be discovered.

9.1 Epistemic types

Six types are used. The taxonomy is specific to this paper's citation profile. The ordering runs from least to most conditional.

- *framework*: a theoretical apparatus the paper adopts and extends. Downstream constructs are conditional on the framework being characterized correctly, not on any single empirical claim.
- *concept*: a specific named idea or result borrowed for one move in the argument.
- *infrastructure*: an existing system referenced as evidence that a described practice exists and has the stated properties.
- *precedent*: a prior instantiation of something this paper proposes. These are cited so that a proposal is not mistaken for an invention, and each carries what the prior instance demonstrated and where its scope falls short of the proposal.
- *illustration*: an empirically grounded phenomenon used to motivate or illustrate, which the architecture does not logically depend on.
- *anchor*: a work the paper responds to. The paper's motivation is conditional on an anchor's diagnosis; its architecture is not.

9.2 Main table: framework

Reference	Identifier	Use in this paper
Ashby (1956)	book; no DOI	Law of requisite variety, the paper's organizing principle: generation variety overtakes regulator variety, and the architecture closes the gap by attenuating one and amplifying the other (abstract; Thesis and Plan; Shore; Kuhn's Diagnosis Through Ashby).
Conant and Ashby (1970)	doi to confirm	A regulator of a system must model that system; the theory atlas is that model made institutional, which is why the cartography layer is load-bearing rather than ornamental (Theory Atlas; Atlas as the Field's Self-Knowledge). Scope note: the theorem's scope is contested in the recent literature, and the paper invokes it in the qualified form "a regulator requires a model, in Conant and Ashby's sense" rather than as settled law.

Reference	Identifier	Use in this paper
Popper (1972)	book; no DOI	World 3 as accumulation through elimination; the Compost Pile operationalizes the negative record and the atlas the positive (Science as Distributed Contraction; Compost Pile; Atlas as the Field's Self-Knowledge).
Kuhn (1962)	book; no DOI	Paradigm crisis given institutional form by the Shore; the architecture separates revolutionary substance from institutional catastrophe (Shore; Monotonicity and Revision; Kuhn's Diagnosis Through Ashby). See Apparent conflicts.
Beer (1972)	book; no DOI	VSM System 5 as the model for the Shore's recursive reassessment of foundational commitments (Shore).
Lakatos (1970)	doi to confirm	Hard-core and protective-belt structure that the trellis extends, with the protection rule deliberately inverted (Trellis). See Apparent conflicts.
Lakatos (1976)	book; no DOI	Proof-generated concepts and the dialectic of conjecture, counterexample, and monster-barring; the Compost Pile and the Shore are that dialectic given institutional memory, with reactivation conditions in place of informal recollection (Compost Pile; Shore).
Ostrom (1990)	doi to confirm	Design principles for institutions governing a rival, subtractable common-pool resource, applied to the field's judgment: Principle 3 for the self-adopted trellis and for free-tier governance, Principle 4 for monitoring accountable to the field, Principle 8 for nested atlas regions (Thesis and Plan; Theory Atlas; A Sorry on the Repository's Governance; Journals in Curation's Purified Form). Scope note: the case record is small-to-medium-group governance of rival natural resources, and the transfer is claimed on the resource having the same structure rather than on the cases covering this one.
Polanyi (1966)	book; no DOI	Tacit knowing, the philosophical home of the smell test; the atlas is specified as the apparatus that converts tacit field-knowledge into explicit queryable form, which is what the amplification half of the Ashby move requires (The Nose; Trellis; Atlas as the Field's Self-Knowledge).
Merton (1973)	book; no DOI	The reward system of science and its norms; the source for what Game 2 is and for the claim that its allocation of credit is a distinct institution from its verification of claims (Thesis and Plan; The Cooperative Equilibrium and the Game 2 Objection; Appendix A).
Dasgupta and David (1994)	doi to confirm	The economics of open science, in which priority is the currency and disclosure is the price; cited for the free-tier claim, since it documents that reward allocation is a plural institutional arrangement rather than a mechanism with a single correct form (Thesis and Plan; The Cooperative Equilibrium and the Game 2 Objection).
MacKenzie (2001)	book; no DOI	The sociology of mechanized proof and the relocation of trust to kernels, compilers, and hardware; cited where the paper bounds its own producer-invariance claim (Commitment 1; The Principle; Thesis and Plan). See Apparent conflicts. This is the second-strongest live objection in the bibliography.

Reference	Identifier	Use in this paper
Bueno and Colyvan (2011)	doi:10.1111/j.1468-0068.2010.00772.x	The inferential conception of applied mathematics and its immersion step; the grounding moment of Section 6 is an immersion event, and naming it places the paper's cross-disciplinary mechanism inside an existing account rather than proposing a new one (The Grounding Moment).
Pincock (2012)	doi to confirm	The mapping account, the rival framework to the inferential conception; cited so that the grounding moment's placement is stated as a choice between live options rather than as the only available reading (The Grounding Moment).
Cartwright (1983)	book; no DOI	Phenomenological departure of specific systems from the certified structures that model them; the departure compost piles are a record of exactly this, given institutional form and a return channel (The Departure Compost Piles).
Mancosu (2008)	doi to confirm	The philosophy of mathematical practice as a field; cited once to place the paper inside the conversation whose methods and questions it uses (Thesis and Plan).

9.3 Main table: concept

Reference	Identifier	Use in this paper
Mayo (2018)	doi:10.1017/9781107286184	Severity; the type-checker as a test of maximal severity for any inconsistency expressible in the type theory (Commitment 1).
Thurston (1994)	doi:10.1090/S0273-0979-1994-00502-6	The narrative structure a proof carries beyond formal correctness: the content the smell perceives and the documentation onus preserves (The Nose; The Onus extends to documentation; Atlas as the Field's Self-Knowledge).
Wigner (1960)	doi:10.1002/cpa.3160130102	The unreasonable effectiveness of mathematics, the recognized ground for the atlas as cross-disciplinary infrastructure, and the question the compression characterization sharpens without resolving (The Atlas as Shared Infrastructure; Mathematics as Certified Compression).
Tanswell (2015)	doi:10.1093/philmat/nkv008	The dependence of an informal result on its formalization is a false friend, since one informal result maps to many formal derivations with no canonical choice; cited for the misstatement failure mode the architecture concedes and routes to the Shore (Modular Axioms and Conditional Formalization). See Apparent conflicts.
Cook (1971)	doi to confirm	Cited for one half of the verification asymmetry only. Checking a given certificate is cheap and that is not conjectural; that finding certificates is expensive in general is the conjecture, and it is what makes the onus a real cost to the generator rather than a formality (Commitment 1; The Principle).
Steiner (1978)	doi to confirm	Mathematical explanation, the literature in which a correct but non-explanatory proof is already a recognized category; the interest

Reference	Identifier	Use in this paper
		crisis is that category under generation pressure (The Two Crises). Scope note: the paper borrows the category and takes no position in the explanation debate.
D’Alessandro (2025)	doi:10.1017/epi.2023.54	Corrigibility as the standard acceptable proofs meet, in place of transferability; cited where the smell test’s tolerance for fixable error needs a criterion (The Two Crises; Generators without Epistemic Agency). See Apparent conflicts.
Star and Griesemer (1989)	doi to confirm	Boundary objects, the literature address for the atlas functioning across two communities that share no vocabulary (The Grounding Moment).
Galison (1997)	book; no DOI	Trading zones and interlanguage, the literature address for the bilingual practitioner the grounding moment requires; cited at the declared sorry so the gap is located in an existing conversation rather than opened fresh (The Grounding Moment).
Kapur (2008)	doi to confirm	Productive failure: struggle before instruction produces transferable understanding in structured domains. Cited as the empirical backing for the pedagogical claim that direct encounter with the block is what trains the operator, together with the boundary condition that consolidation must follow the struggle (Pedagogical Consequences).
Pnueli (1985)	doi to confirm	Assume-guarantee reasoning: a component is verified against assumptions its context must discharge. The modular axiom discipline is this pattern applied to mathematical papers, with the discharge event supplied by library formalization (Modular Axioms and Conditional Formalization).

9.4 Main table: infrastructure

Reference	Identifier	Use in this paper
de Moura and Ullrich (2021)	doi to confirm	Lean 4 and dependent type theory; trellis constraints as typeclasses, hypothesis-testing as compilation, and the source of the term *sorry*, which this paper adopts as its governance vocabulary for a characterized gap (Trellis; Compost Pile).
The mathlib Community (2020)	doi:10.1145/3372885.3373824	Mathlib style conventions as the documentation practice the onus extends, and Mathlib as a verified library with a typeclass-hierarchy chart (The Onus extends to documentation; Theory Atlas; Cooperative Equilibrium).
Paulson, Nipkow, and Wenzel (2019)	doi to confirm	Isabelle/HOL and its theory-graph chart as one of the libraries the atlas vocabulary generalizes (Theory Atlas). Scope note: cited in body for Isabelle’s Archive of Formal Proofs; the source documents Isabelle/HOL, and the AFP is a related but distinct artifact.

Reference	Identifier	Use in this paper
Bancerek et al. (2018)	doi to confirm	The Mizar Mathematical Library as one of the community-maintained verified libraries (Theory Atlas; Cooperative Equilibrium).

9.5 Main table: precedent

Reference	Identifier	What it instantiates, and where its scope stops
Hales et al. (2017)	doi:10.1017/fmp.2017.1	The onus of formalization already imposed once. The Kepler conjecture proof announced in 1998 and published in 2005 could not be certified by its referees, and the discipline's answer was the Flyspeck project, completed in 2014 and published here. This is the architecture's central principle deployed in response to exactly the verification failure the paper diagnoses (The Principle; Modular Axioms and Conditional Formalization). Scope: one proof, retrospectively, by the original author's initiative, with no institutional rule following from it.
Hales (2024)	arXiv:2402.08032	The practitioner's critical reappraisal of that project: what formalization cost, what it delivered, and what it did not. Cited so the Flyspeck precedent enters with its own author's assessment rather than as an unqualified success (The Principle). Scope note: preprint, and a self-assessment by the project lead.
Gonthier et al. (2013)	doi to confirm	Mechanized verification at research scale before AI generation, demonstrating that the onus is payable on a major theorem (The Principle).
Necula (1997)	doi to confirm	The onus principle in computer security: the producer ships a proof, the consumer checks it cheaply, and trust relocates to the checker. The architecture proposed here is proof-carrying code applied to mathematical publishing (The Principle; Commitment 1). Scope: proof production was the bottleneck that limited adoption, which is the paper's own deployment risk observed in a prior deployment, and the reason the modular axiom discipline is load-bearing rather than convenient.
Boyer et al. (1994)	doi to confirm	The QED Manifesto: the program of a mechanized encyclopedia of mathematics, stated thirty years before this paper and largely unrealized in its original form (Theory Atlas). Scope: QED proposed the library and did not specify the institutional arrangement that would produce it, which is where this paper's contribution sits.
Avigad and Harrison (2014)	doi to confirm	The state of formally verified mathematics at the point the current libraries became viable; cited for the intellectual history the architecture rides rather than originates (Theory Atlas; Cooperative Equilibrium).
Voevodsky (2014)	grey literature	A Fields medalist reaching this paper's diagnosis from inside the profession: that informal verification fails at the scale of his own work, and that the response is formalization (Generators without Epistemic

Reference	Identifier	What it instantiates, and where its scope stops
		Agency; Cooperative Equilibrium). Scope note: an institute newsletter, cited for a documented personal account rather than as a study.
National Academies (2014)	doi to confirm	Community-owned structural infrastructure over the mathematical literature, proposed a decade before this paper. The theory atlas is a successor proposal and not a first one (Theory Atlas). Scope: the report proposed the library and left the admission criterion unspecified, which is what the onus supplies.
Gowers and Nielsen (2009)	doi to confirm	Polymath as a demonstration that distributed mathematical work coordinates without institutional realignment; the empirical precedent for the coordination function attributed to repositories (Repositories as Verifiers, Cartographers, Coordinators; Cooperative Equilibrium).
Moravcsik and Murugesan (1975)	doi to confirm	Citation typing by function, fifty years prior: conceptual against operational, organic against perfunctory, confirmatory against negational. The precedent for this appendix's own discipline (Appendix B). Scope: reconstructed by analysts after publication, where this appendix places the obligation on the author before admission.
Shotton (2010)	doi to confirm	CiTO, typed citation implemented as a machine-readable ontology; the working precedent for the reference atlas's typed edges (The Reference Atlas; Appendix B). Scope: CiTO types edges an author declares, where the reference atlas computes the neighborhood from content, which is the whole of the difference. Related note: zbMATH Open and MathSciNet are the existing curated attribution layer over the mathematical literature and are the reference atlas's nearest working relatives.

9.6 Main table: illustration

Reference	Identifier	Use in this paper
Rosenthal (1979)	doi:10.1037/0033-2909.86.3.638	The file-drawer problem as the pathology of lost negative knowledge that the Compost Pile remedies, the zombie-iteration motivation (Science as Distributed Contraction). Scope note: this is also the boundary condition on the Compost Pile, since a restriction whose discarded class was never observed cannot be converted into a named entry.
Fanelli (2012)	doi to confirm	The quantitative wing of the same phenomenon: negative results declining as a share of the published record across disciplines and countries (Science as Distributed Contraction).
Krakovna et al. (2020)	blog; grey literature	Specification gaming as the mechanism by which a reward-maximizing generator produces odorless proofs of the first kind (Generators without Epistemic Agency).
Andersen (2020)	doi to confirm	Interview evidence on what referees actually check and knowingly skip; the empirical grounding for the paper's claim that structural flaws persist when reviewer effort is bounded (The Principle). See Apparent conflicts.

Reference	Identifier	Use in this paper
Greiffenhagen (2023)	venue and doi to confirm	A second empirical study of correctness-checking in mathematical peer review, cited alongside Andersen so the bounded-effort premise rests on more than one study (The Principle).
Greenberg (2009)	doi to confirm	Citation distortion demonstrated on a real citation network: claims acquiring unfounded authority through the gap between the declared citation graph and the graph the corpus exhibits. This is the declared-versus-computed delta shown empirically rather than argued (The Reference Atlas; Dependency Chains and Citogenesis).
Tymoczko (1979)	doi to confirm	The four-colour theorem as the discipline's first philosophical crisis over a proof no human could survey; the precedent for treating surveyability and mechanical checking as a question about what a proof is for (Commitment 1; Commitment 2).
Scholze and Stix (2018)	grey literature; hosted copy to confirm	The abc case, cited as the highest-profile recent operation of the smell test and as the worked instance of the paper's separation thesis: priority settled by date and content while the correctness verdict remains unsettled, with a journal's curatorial decision decoupled from the field's verification verdict (The Two Crises; Journals in Curation's Purified Form; The Reference Atlas). Scope note: an unrefereed report in an unresolved dispute, cited for the institutional event rather than for a verdict on the mathematics.
National Academies (2019)	doi to confirm	Reproducibility and replicability as the frame funding bodies already operate in; cited for the claim that funders face accountability pressure the architecture answers (Cooperative Equilibrium).
Wu et al. (2022)	doi to confirm	Autoformalization by language models, cited for the specific pressure the premise names: generation reaching the formalization bottleneck itself (Thesis and Plan; Generators without Epistemic Agency).

9.7 Main table: anchor

Reference	Identifier	Use in this paper
Klowden and Tao (2026)	arXiv:2603.26524	The smell test, the odorless proof, the blue-team and red-team distinction, and the citogenesis case: the diagnosis the paper explains and the phenomena it addresses (The Nose; The Onus extends to documentation; Red-Teaming as Competitive Function; Dependency Chains and Citogenesis). See Status watch.
Jaffe and Quinn (1993)	doi to confirm	The prior round of the discipline's public argument over what verification costs and who bears it; cited so the AI moment reads as the third round of an existing fight rather than as an unprecedented event (The Two Crises; Thesis and Plan).

Reference	Identifier	Use in this paper
Responses to Jaffe and Quinn (1994)	doi to confirm	The sixteen-respondent exchange, including the response this paper already cites as Thurston (1994); cited so that Thurston's essay enters with its occasion visible rather than as a free-standing statement (The Nose; The Two Crises).
DeMillo, Lipton, and Perlis (1979)	doi to confirm	The claim that mathematical confidence is produced by social process rather than by formal derivation, and that program verification lacks the social process that makes mathematics work. The direct ancestor of this paper's frame and its strongest structural objection (Thesis and Plan; Commitment 1). See Apparent conflicts.

9.8 Apparent conflicts

Works cited in this paper that appear to contradict its conclusions, each with a typed resolution. This is the section that requires understanding why two positions do not actually conflict, or where the paper concedes that one partly does.

Cited work	Apparent conflict	Resolution type	Resolution
Lakatos (1970)	Lakatos protects the hard core by methodological decision, so the core is shielded from refutation by convention; the trellis claims every constraint is revisable on evidence.	explicit-departure	The paper adopts the core and belt structure and inverts the protection rule, stated in the body as "against Lakatos." The conflict is declared, not overlooked: the trellis is a Lakatosian program with a conjectural rather than conventional core.
Kuhn (1962)	Kuhn binds revolutionary substance to incommensurability and institutional catastrophe; the paper claims the two separate.	scope-difference	Kuhn documented disciplines whose institutional apparatus lacked requisite variety. The argument is that the binding he observed is contingent on that missing apparatus. When the regulator's variety matches the change, substance and catastrophe come apart. The architecture supplies the apparatus, so Kuhn's documentation stays accurate for the disciplines he studied.
Tanswell (2015)	Tanswell argues informal proofs do not reduce cleanly to formal ones, since a single informal result corresponds to many formal derivations with no canonical choice among them. This reads as an objection to an architecture that treats the type-checking proof term as the authoritative record.	acknowledged-limitation	The objection lands in part. The architecture does not claim the formal term captures the full informal content; it claims the formal term is what can be mechanically verified, and the documentation onus carries the informal narrative alongside it. The residual gap, a misstatement between an informal result and its chosen formalization, is exactly the failure mode the paper concedes: it stays invisible until a sufficiently different formalization attempt exposes it, and the paper routes it to the Shore and to accumulating discharge evidence rather than

Cited work	Apparent conflict	Resolution type	Resolution
			claiming immunity. This is the strongest live objection in the bibliography and the resolution is a partial concession, not a dismissal.
DeMillo, Lipton, and Perlis (1979)	Their argument is that what makes a mathematical proof believable is the social process it survives, that formal derivations are unread and unreadable, and that a discipline which replaces social process with formalization loses the thing that was doing the work. The architecture proposes precisely that replacement.	scope-difference-plus-concession	Two of their three premises have been overtaken by events rather than refuted by argument, and the paper should say which. Formal derivations are no longer unread, because proof assistants read them, and the libraries they read have grown to research scale, which Gonthier et al. (2013) and Hales et al. (2017) demonstrate. What survives intact is their central point, and the architecture agrees with it: social process is what confers *interest*, and the paper assigns interest to curation and to the smell rather than to the verification layer. The disagreement is narrower than it appears and falls entirely on correctness. Where they claim social process is the only available mechanism for confidence in correctness, the paper claims a mechanical one is now available and cheaper. Their prediction that formalization would not scale was reasonable in 1979 and is the specific claim the intervening record has answered.
MacKenzie (2001)	The paper claims a kernel's verdict is producer-invariant and that verification is therefore zero-trust. MacKenzie's cases show that mechanized proof relocates trust rather than eliminating it: kernels contain bugs, compilers and hardware are trusted implicitly, and disputes over what a machine-checked proof establishes are social disputes of the kind the architecture claims to have mechanized away.	acknowledged-limitation	The objection lands and the paper's claim is bounded accordingly. Standing is invariant under variation of the producer, which is the attenuation Section 1 defines, and it is relative to a checker, which the definition does not remove. What the architecture buys is that the residual trust is concentrated in one small artifact that many parties independently reimplement and recheck, which is the de Bruijn criterion, rather than distributed across the bounded attention of referees. Relocation to a small independently rechecked kernel is a different and better position than the one MacKenzie's cases criticize, and it is a weaker claim than zero trust. The paper claims relocatability and auditability of trust rather than its abolition.
Andersen (2020)	Andersen documents that mathematicians knowingly accept gaps in proofs they referee, and that this is competent practice rather	scope-difference	The finding supports the paper's premise and corrects its framing. Referees skipping steps is evidence that reviewer effort is bounded and allocated by judgment, which is the paper's diagnosis, and the architecture's response is to move

Cited work	Apparent conflict	Resolution type	Resolution
	than failure. The paper treats unchecked structure as the pathology its architecture removes.		the certifiable part off the referee's budget rather than to demand that referees check more. Where Andersen's finding bites is the implication that gap-acceptance is a skill rather than a shortfall, and the paper accepts that: what remains after mechanization is the judgment that decides which unformalized things are worth accepting, which is the smell, and the architecture preserves it rather than replacing it.
D'Alessandro (2025)	Acceptable proofs need only be corrigible, and proofs containing substantive fatal errors can be acceptable when experts can fix them. An admission gate that rejects anything failing to type-check enforces a standard the discipline does not actually hold.	scope-difference	The two standards govern different objects. Corrigibility is a standard for what the community should engage with, which is the proposal layer, and the paper's gate is a standard for what enters the verified record, which is the acceptance layer. A broken but fixable proof is a good submission and an inadmissible record entry, and the architecture has a place for it: the Compost Pile entry, the declared axiom, the characterized gap. The conflict would be real if the paper claimed unadmitted work is worthless, and it claims the opposite, since negative and conditional records are its primary deliverable.

9.9 Status watch

Klowden and Tao (2026), arXiv:2603.26524v1, is fresh and pre-peer-review. If the diagnosis is substantially revised, the paper's motivating frame shifts, but the architecture is an institutional-design argument that does not take the diagnosis as a premise, so the architecture is unaffected. If the diagnosis is confirmed at peer review, the smell-test and odorless-proof framing promotes from one team's coinage to an established characterization, which strengthens the opening without further edits.

Hales (2024) is a preprint and a self-assessment by the project it assesses. It is cited for the author's own reported judgment of the Flyspeck effort and not as an independent evaluation of it.

Scholze and Stix (2018) is an unrefereed report in a dispute that has not resolved. It is cited for the institutional event, which is documented, and carries no position on the underlying mathematics.

Greiffenhagen (2023) carries an unconfirmed venue and volume and must be resolved before submission. Every entry above marked "doi to confirm" is an identifier this paper has not yet verified against a catalog, and each is a declared gap of the same kind the modular axiom discipline names: a placeholder in use, flagged, with discharge pending. An entry that cannot be resolved is withdrawn rather than softened.

This appendix currently runs ahead of the body. Each row states the use its reference is admitted for, and a use that the body does not yet make is an obligation on the next revision rather than a claim about the present text. Any row whose stated use goes undischarged is withdrawn, on the same rule the appendix applies to a reference with no traceable use.

9.10 Acyclic dependency

None required. This paper cites no companion or concurrently circulated work in its bibliography, and it names no forthcoming piece as the place where an obligation is discharged. Every construct the paper relies on is either specified in the body or held open as a declared sorry here, so there is no in-paper citation cycle to disclaim and no deferral resting on an unpublished source.